HURRICANE READINESS PROJECT

TEAM 4

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Overview

The Steering Committee of the Center for Risk Management (hereafter the "Committee") seeks to better understand the expected rate of failure for pump operations in the gulf area. When hurricanes hit this area, pumps are at risk of flood-related damage or mechanical failures, which render the machine inoperable. The Committee is particularly concerned about the impact of runtime on motor-related failures.

To address these concerns, our team built a Cox proportional hazard (PH) model for motor failures. We found that the most significant predictors of these failures are the pump's age and the slope of the surrounding ravine. Further analysis demonstrated that running a pump for 12+ hours does not impact failure risk. Based on our findings we recommend that the Committee pay close attention to newly installed pumps, which have a higher risk of motor failure.

Methodology

This section describes the data used for modeling, including any necessary modifications.

Data Used

Our team analyzed a training set with 770 observations, with each observation representing one pump. Pump status *(i.e., operational or failed)* was recorded for each hour of the critical 48-hour emergency period. A separate variable recorded the number of hours until the pump either failed or was censored from the data. Pump attributes were organized into nine additional predictors.

Data Preparation

All pumps have a listed reason for failure, labeled from 0 *(i.e., no failure)* to 5. Our analysis focused on the motor failure type (represented by 2 in the reason column). We therefore created an updated target variable to record whether a pump experienced a motor-related failure *(target equals 1)* or not *(target equals 0)* during the 48 hour period.

We binned the pump's age variable into four quartile categories, with the first category representing the youngest pumps and the fourth category representing the oldest pumps. We binned the slope variable into three tertile categories. This modification removed the linearity assumption required for continuous variables, since we were able to treat age and slope as categorical. Our team treated all other numeric variables as categorical.

Finally, to determine if constantly running a motor for 12+ hours impacts its risk of failure, we transformed the data into long format. In this format, each row contains information about one pump over a one-hour period. We then created a flag variable that indicates if the motor has been running for at least 12 consecutive hours.

Analysis

This section describes the modeling process to explore the hazard rate of motor failure.

Cox Hazard Model

To align with the Committee's current variable selection strategies, our team used backward selection to eliminate insignificant predictors of motor failure. In this scenario, a model with fewer predictors is more desirable, as it would simplify any Committee decisions regarding budget allocation for pump upgrades. Therefore, our team used the Bayesian Information Criterion (BIC) criteria for the selection process, which penalizes more complex models. We also confirmed that both variables satisfied the proportional hazard assumption by running the appropriate test on the Schoenfeld residuals.

Table 1 lists the variables in the final Cox PH model, along with their description and p-value. Variables are listed in descending order of significance based on p-value.

Table 1: Final Cox-PH predictors for motor failure, selected by backwards elimination.

Variable	Description	P-Value
Age (Q3)*	Describes pumps from ages 7.1 to 7.8.	8.84×10^{-13}
Age (Q4)*	Describes pumps from ages 7.9 to 10.4.	1.47×10^{-11}
Age (Q2)*	Describes pumps from ages 6.6 to 7.	4.88×10^{-11}
Slope (T3)**	Indicates that the surrounding slope ranges from 3ft to 18ft.	5.81×10^{-6}
Slope (T2)**	Indicates that the surrounding slope ranges from 2ft to 3ft.	0.967

**All original ages are the difference between the pump's installation date and the current date. Age reference level was Q1, which contains pumps from ages 5.3 to 6.5.*

***All original ravine slopes were assumed to be measured in feet. Slope reference level was T1, which describes a surrounding ravine slope ranging from 0ft to 2ft.*

As seen in Table 1, the most significant feature contributing to a pump's hazard rate was its binned age, with all comparisons made to the youngest pumps. For example, we found that for pumps between 7.1 and 7.8 units old, the hazard rate of motor failure decreased by **87.84%** compared to pumps between 5.3 and 6.5 units old, holding all other variables constant. In general, at any given point during the hurricane, an older pump has a lower rate of motor failure than a younger pump *(ignoring the effects of other factors)*. Finally, while the indicator for surrounding slopes ranging from 2ft to 3ft was not statistically significant, it was retained because its counterpart for slopes ranging from 3ft to 18ft *was* statistically significant.

When building this model, our team also discovered that the indicator for trash rack (a piece of hardware that protects the pump from debris) had quasi-complete separation with the target variable. In other words, none of the 398 pumps protected by a trash rack experienced a motor failure. Including this variable in a statistical model would result in an uninterpretable coefficient due to non-convergence. However, this separation may suggest that protecting a pump with a trash rack greatly reduces the risk of motor failure. It is important to note that the addition of a trash rack does not necessarily reduce the pump's risk of failing for non-mechanical reasons.

Motor Runtime Impact

The Army Corp of Engineers and the Committee were interested in assessing the risk of motor failure for pumps that constantly ran for 12 hour blocks. We transformed the data so that each row recorded the status of each pump for one hour during the critical 48 hour time period. Status was equal to 1 if the pump was running, and 0 otherwise. Pumps that never suffered a motor failure were thus represented by 48 rows; all other pumps stopped being recorded after the hour that they failed. We flagged each 12+ hour block of consecutive runtime to use as a time-dependent predictor of motor failure.

Finally, we built a new Cox PH model using the pump's binned age, the binned slope of the surrounding ravine, and the pump's runtime. This new data format allowed for variable effects that may change over time. We found that the runtime variable was insignificant at the $\alpha=0.03$ level (p=0.480), while the binned age and slope variables still significantly impacted hazard rate.

Results & Recommendations

Based on these analyses, we recommend that the Committee focus on assessing pumps based on their age and the surrounding ravine slope, if they intend to reduce the specific risk of motor failure. The pumps at lower risks of this failure are those that are older and/or stationed in areas with a larger slope value. This is why we suggest the Committee investigate newly installed pumps to look for a reason that those pumps are more prone to motor failure.

Future research should focus on exploring the relationship between other pump failure reasons and age, ravine slope, and trash racks. Our analysis shows significant relationships between age and motor failure and ravine slope and motor failure, but does not prove any causal relationships.

Conclusion

By identifying the significant relationships between motor failures and certain pump characteristics, such as its age category, this analysis will inform the Committee's strategy to improve pump performance. Data shows that younger pumps and those situated on minimal slopes are at higher risk of motor failure; in response, the Committee is encouraged to closely monitor new pumps, and prioritize steeper slopes for installation.

If the Committee's budget allows for additional upgrades, our team believes that new trash rack installations would be the most immediately beneficial modification. By implementing these recommendations, the Steering Committee of the Center for Risk Management will be better prepared for future hurricane seasons.